PRECISION-GNSS.COM · INTERNAL PROJECT DOCUMENT

GNSS AI Translation Engine

An executive summary of what the system does, how it works, and what it costs to run — English → Spanish technical translation, built for accuracy.

● Translation Engine complete · WordPress integration pending staging access

INTRODUCTION

What this project is, in plain words

Precision-GNSS.com currently exists only in English. This project builds a small, dedicated piece of software whose only job is to read the site's articles, translate them into natural, technically accurate Spanish, and hand that translation back ready for a person to approve — before anything ever goes live.

Think of it less like a "translate this page" button, and more like hiring a careful bilingual technical editor who never gets tired: it reads each piece of content, writes a translation, then reads its own work a second and third time looking specifically for mistakes — a wrong number, a softened warning, an inconsistent technical term — before ever passing it along. Nothing is published automatically. A human always has the final say.

The reason this matters is that the content in question is technical: measurements, tolerances, product specifications, safety notes. A generic translation tool optimises for text that reads smoothly. This system is built to optimise for something else first — that the Spanish version says *exactly* the same thing as the English original, down to the last centimetre.

3
Independent AI checks per translation

60
Automated tests, all passing

<$0.01
Typical cost per translated page

0
Pages ever published without review

How a translation actually happens

INPUT
Source text
+ surrounding context

→

STEP 1
Translator
writes the Spanish version

→

STEP 2
Technical Reviewer
checks nothing changed meaning

→

STEP 3
Terminology Check
confirms correct vocabulary

Only after passing all three steps does a translation become a draft, waiting for a human to publish it.

UNDER THE HOOD

Technology & Architecture

The system is deliberately built as three separate pieces that each do one job well, rather than one large tangled script. WordPress never talks to the AI directly, and the AI never has any awareness of WordPress — a dedicated engine sits between them and owns all the logic, rules, and record-keeping.

WEBSITE
WordPress
WPML · Elementor · Yoast

→

BRIDGE
gnss-bridge
small connector plugin

→

CORE
Translation Engine
Python — implemented

→

AI
DeepSeek API
language model

The connector plugin exists because WordPress's translation plugin, WPML, does not offer a public way to create translations from outside WordPress — that can only be done from code running inside WordPress itself. A small, auditable bridge is the safest way to reach it, instead of writing directly into WPML's database, which its own documentation advises against.

Technology stack

LAYER	TECHNOLOGY	ROLE
Translation engine	Python 3.10	Independent service, outside WordPress
AI provider	DeepSeek API	`deepseek-v4-pro` / `deepseek-v4-flash`, JSON-structured output
Data validation	Pydantic v2	Every AI response is strictly type-checked
Testing	pytest	60 automated tests, no live calls in CI
Configuration	python-dotenv	Secrets stay out of source control
CMS integration *(designed)*	WordPress REST API	Standard, authenticated content access
Translation memory *(designed)*	SQLite	Only re-translate what actually changed

HOW IT THINKS

The Translation Pipeline, in Detail

Translation and quality-checking are kept as separate, independent AI calls rather than one combined request — and that choice has already proven itself. During testing, the Translator alone produced a subtle but real technical error: it translated "*achieves a fix within 5 seconds*" (meaning *at most* 5 seconds) as "*en menos de 5 segundos*" (meaning *strictly less than* 5 seconds). The independent Reviewer step caught this exact error and correctly rejected the translation. A single combined request would very likely have missed it.

1 · Translator

Writes the Spanish translation under a strict rulebook: preserve exact meaning, numbers, units, conditions and warnings; never simplify, invent, soften, or drop technical information; never translate protected product names. Returns the translation together with a confidence score.

2 · Technical Reviewer

Independently compares the original and the translation side by side, specifically hunting for information that was added, removed, or subtly altered — exactly the kind of error shown above.

3 · Terminology Validator

Checks the translation against a curated glossary of GNSS/RTK/surveying terms, flagging any case where required vocabulary (e.g. "*base station*" → "*estación base*") wasn't used correctly.

Glossary & long-content handling

Only the glossary terms relevant to the current text are sent to the AI, keeping every request small and inexpensive. Long articles are split at paragraph boundaries — never mid-sentence, and never mid-number — so a figure like "10 km" is never cut apart, translated separately, and reassembled wrong.

WHERE THINGS STAND

Implementation Status

The project is organised into ten phases. The core translation engine is complete and validated; everything that touches the live WordPress site is designed but waiting on staging access.

PHASE	SCOPE	STATUS
0	WordPress/WPML environment audit	● Needs staging access
1	gnss-bridge connector plugin	● Designed, not deployed
2	WordPress connector (Python)	● Needs staging access
3	Content extraction (Elementor/Yoast)	● Needs staging access
4	Translator, Reviewer, Terminology Validator, chunking	● Complete — real-API validated
5	Glossary engine	● Complete — seed glossary
6	Translation memory & change detection	● Not started
7	QA scoring engine	● Not started
8	WPML orchestration, end to end	● Needs staging access
9	Pilot on 5 real pages	● Needs staging access
10	Scale-out & scheduled sync	● Not started

Full detail is tracked in `ROADMAP.md` , `PLA-ACCIO.md` and `LOG.md` in the project repository.

WHAT IT COSTS

Cost Analysis

Rather than estimate costs theoretically, four real, public pages were pulled from the live site and run through the engine's actual content-splitting logic and real prompt sizes — the figures below are measured, not guessed. DeepSeek bills in USD; EUR figures alongside are converted at **$1 = €0.876** (23 July 2026 reference rate) for convenience.

$0.0064
Avg. cost per page — translation only (€0.0056)

$0.0129
Avg. cost per page — with full review (€0.0113)

$0.00085
Per 1,000 characters — full pipeline (€0.00074)

DeepSeek pricing (as of 23 July 2026)

MODEL	INPUT	OUTPUT	USED FOR
`deepseek-v4-pro`	$0.435 (€0.381) / 1M tokens	$0.87 (€0.762) / 1M tokens	Translation (default)
`deepseek-v4-flash`	$0.14 (€0.123) / 1M tokens	$0.28 (€0.245) / 1M tokens	Optional, cheaper QA model

Real pages tested

PAGE	WORDS	TRANSLATE ONLY	FULL REVIEW PIPELINE
Homepage	1,800	$0.00523 (€0.00458)	$0.01049 (€0.00919)
Archaeology (RTK application)	2,156	$0.00585 (€0.00512)	$0.01173 (€0.01028)
Precision Agriculture (RTK application)	3,951	$0.01040 (€0.00911)	$0.02083 (€0.01825)
News — RTK/GNSS for Robotics	1,575	$0.00425 (€0.00372)	$0.00852 (€0.00746)

Worked example — one page, step by step

Take the *Precision Agriculture* page above: **24,706 characters** of translatable prose, measured directly from the live page. Applying the flat rate of **$0.00085 per 1,000 characters** (full review pipeline, `deepseek-v4-pro`) gives:

24,706 chars × $0.00085 ÷ 1,000 = **$0.0210 (€0.0184)** — matching the $0.02083 measured directly from the real API call above (the small difference is normal token-count rounding). Translation only (no independent review) comes to about **$0.0104 (€0.0091)** for the same page — roughly half, since it skips the Reviewer and Terminology Validator calls.

In practical terms: translating and fully reviewing a typical content page costs about **one-hundredth of a dollar (about one cent in euros too)**. A complete first pass across 50 comparable pages would cost roughly **$0.64 (€0.56)** in total. Ongoing costs after launch are far lower still, since only content that actually changes ever needs to be re-translated.

WHAT HAPPENS NEXT

Current Status & Next Steps

Done

- Translation Engine core — Translator, Reviewer, Terminology Validator, long-content handling.
- Glossary Engine — curated domain terminology, applied automatically per translation.
- 60 automated tests, plus real validation against the live DeepSeek API and live site content.

Blocked — waiting on staging access

- WordPress connector, content extraction, and the full end-to-end publishing pipeline.

Available to start now

- QA scoring engine (numbers, units, URLs, protected terms).
- Translation memory, so unchanged content is never re-translated.
- The connector plugin skeleton — can be written and reviewed without being deployed.